From:	richard carchman [walntz@msn.com]
Sent:	Wednesday, October 10, 2007 10:23 AM
To:	Fisher, Michael T.
Cc:	Podraza, Ken F.
Subject:	RE: tsna draft commentary

Confidential

I understand that. I can only **deal** with a scientific document that provides the data and logic for such a decision and not a web or press release.... But I have forwarded this to Ken as well. thanks

Richard

Subject: RE: tsna draft commentary
Date: Wed, 10 Oct 2007 10:14:07 -0400
From: Michael.T.Fisher@pmusa.com
To: walntz@msn.com; Ken.F.Podraza@pmusa.com

Thanks Richard,

NNN and NNK are currently listed as Group I human carcinogens by IARC as shown below in the mention pasted off the IARC website.

N'-Nitrosonornicotine (NNN) [16543-55-8] and 4-(*N*-Nitrosomethylamino)-1-(3-pyridyl)-1-butanone (NNK) [64091-91-4] (Vol. 37, Suppl. 7, Vol. 89; in preparation)
(NB: Overall evaluation upgraded from 2B to 1 based on mechanistic and other relevant data)

Michael

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From: richard carchman [mailto:walntz@msn.com]
Sent: Wednesday, October 10, 2007 9:29 AM
To: Podraza, Ken F.
Cc: Fisher, Michael T.
Subject: FW: tsna draft commentary

Hi Ken; I have attempted to distill down to the most basic level my view of the role of TSNA's and human lung cancer. I believe that this is as condensed as I can make it and still make sense. My original attempt at this was ~5 single spaced pages , not including tables and figures. Please let me know if this works? Tony Tricker provided his usual amount of invaluable assistance. For him it was quite easy , since he is one of the worlds experts in this arena.

Richard

From: walntz@msn.com
To: walntz@msn.com
Subject: tsna draft commentary
Date: Wed, 10 Oct 2007 09:19:31 -0400

The Tsna's rose to prominence in ~ 1978 when scientists at the AHF were able to measure them in tobacco and tobacco smoke. This observation forced them to reverse their previous recommendations concerning adding nitrate to tobacco to reduce PAH's., since adding nitrate increased TSNA levels. Currently two of the TSNA's are listed by

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Iarc(1985-vol38) as possible human carcinogens (2b) based on animal data. It appears that IARC may be changing their position by listing them as known human carcinogens. There is no human data available that supports this change. On the contrary , the human data that exists from smokeless (see consensus) and from cigarettes (see Ted) provide no support to these compounds being important for lung cancer in humans. TSNA's as the name implies are tobacco specific nitrosamines. They bear a structural relationship to nicotine. This structural relationship to nicotine has two implications. The first is that since TSNA's are not toxic per se. They have to be metabolized to become active and these active compounds may be further metabolized to inactive compounds and excreted. (for good reviews see Hecht and Tricker...analytical determination of nicotine and related compounds and their metabolites, elsevier, 1999 and Nilsson.Inter J of Occup med and envir hlth vol19-2006). The importance of the structural similarities to nicotine , shared metabolism (with important differences in the Km and Vmax for these enzymes) and the tremendous differences in the amounts of these materials (can be ~one million times different) in tobacco and smoke will be an important considerations in evaluating the role of TSNA's in humans. The structural similarities to nicotine have also been shown that TSNA's can bind to and activate the nicotinic cholinergic receptor (this will not be discussed further in this commentary). For the sake of space I will only discuss NNK. This compound can produce a variety of cancers in certain animal models via a variety of routes of exposure (ie po , ip sc).I will further only focus on lung cancer with this compound in these animal models. There is a tremendous difference in the sensitivity of these animal models(~1000 fold) the SG hamster is the most sensitive model. When NNK is metabolized to reactive species a number of outcomes are observed , the one that has received the most attention are adducts; either to DNA or to hemoglobin...The most important question following this logic is whether the human , particularly the lung has the same metabolic capacity as the sensitive animal models. Most of the studies that looked at the precise metabolic pathways used tissue homogenates. There are two issues associated with these studies , one is that homogenates lack the normal cell partioning elements (ie membranes) and the specific activity of the NNK substrate was such that the actual amount of NNK impacted which metabolic enzymes were being measured (this is related to differences in the Km and Vmax's of the different metabolic enzymes)... Recent work led by Tricker et al using precision cut tissue slices and very high specific activity substrates (eg nnk) have provided a clearer picture of the metabolic and adduct pattern comparing animals and humans-(eg see Richter et al-.Proc of15th Inter. sympos. microsomes and drug metabolism.2006). What they concluded from their studies using NNK in the mouse , rat , hamster and human lung is that both the metabolite pattern and adduct patterns are significantly different between the animal and human preparations. An important study by the Lovelace group in New Mexico . exposed AJ mice to tobacco smoke with and without NNK. What they found was that tobacco smoke exposure didn't increase lung tumors , NNK did (PO) and the combination of the two showed decreased lung tumors compared to the NNK exposed group. The authors were surprised by these findings..Tricker et al were able to demonstrate that since nicotine competes with NNK for metabolism that both NNK metabolites and adducts were significantly reduced in the presence of nicotine (nicotine is normally present in tobacco and smoke at ~1 million times higher levels) . This then allows the nicotine to effectively compete for these enzymes much better than NNK, ie decreased active nnk metabolites , decreased adducts , decreased lung tumors... The published adduct studies in humans show no clear picture to support the metabolic pathways seen in the sensitive animal models. **Summary**...Some TSNA's are lung tumorigens via different routes of exposure. in animals..Human have been exposed to very high levels of TSNA's (smokeless) for decades and the epidemiology does not support an association with lung cancer . Cigarettes smoke containing different levels of TSNA's show no impact of tsna's in humans (see Ted). The metabolic and adduct profile seen in animals and humans are significantly different. even if there was a NNK in the lung the amount of nicotine would overwhelm any ability for metabolic activation and adduct formation.... At this point , I am not aware of any compelling data regarding TSNA's and lung cancer in humans

Richard

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